

Data Dictionary

Road Accident Analysis Project

Dataset Name

Road Accident Data

Project Title

Road Accident Analysis Dashboard using Excel, SQL and Power BI

Purpose of the Data Dictionary

This data dictionary explains the meaning, data type, source table, and analytical use of each column used in the Road Accident Analysis project. The dataset was cleaned in Excel, imported into MySQL, divided into relational tables, and visualized in Power BI.

Data Dictionary Table

Column Name	Suggested Data Type	Source Table / Category	Description	Use in Analysis
accident_index	VARCHAR / Text	Main Identifier	Unique ID for each accident record	Used as the primary key to identify each accident and connect tables
accident_date	DATE	accident_fact	Date on which the accident occurred	Used for monthly, yearly, and trend analysis
accident_time	TIME	accident_fact	Time when the accident occurred	Used for time-based accident analysis
day_of_week	Text	accident_fact	Day of the week when the accident happened	Used to analyze accident patterns by weekday
accident_severity	Text	accident_fact	Severity level of accident such as Slight, Serious, or Fatal	Used to analyze accident seriousness and risk
number_of_casualties	Integer	accident_fact	Number of casualties	Used to calculate total casualties

Column Name	Suggested Data Type	Source Table / Category	Description	Use in Analysis
			involved in the accident	and average casualties
number_of_vehicles	Integer	accident_fact	Number of vehicles involved in the accident	Used to understand vehicle involvement in accidents
speed_limit	Integer	accident_fact	Speed limit of the road where the accident occurred	Used to analyze accident patterns by speed limit
year	Integer	Derived Column / accident_fact	Year extracted from accident date	Used for yearly filtering and trend comparison
month	Text	Derived Column / accident_fact	Month name extracted from accident date	Used in monthly accident trend charts
month_number	Integer	Derived Column / accident_fact	Numeric month value from 1 to 12	Used to sort month names correctly in Power BI
hour	Integer	Derived Column / accident_fact	Hour extracted from accident time	Used for time-of-day analysis
severity_score	Integer	Derived Column / accident_fact	Numeric value assigned to severity, such as Slight = 1, Serious = 2, Fatal = 3	Used to support severity-based analysis
serious_fatal_flag	Integer	Derived Column / accident_fact	Flag value where 1 means Serious/Fatal and 0 means Slight	Used to calculate Serious/Fatal Accident Rate
latitude	Decimal	accident_location	Latitude coordinate of	Used for geographical or

Column Name	Suggested Data Type	Source Table / Category	Description	Use in Analysis
			accident location	map-based analysis
longitude	Decimal	accident_location	Longitude coordinate of accident location	Used for geographical or map-based analysis
local_authority_district	Text	accident_location	District or local authority where the accident occurred	Used to identify high-accident districts
police_force	Text	accident_location	Police force area responsible for the accident record	Used for regional/police-force-based accident analysis
urban_or_rural_area	Text	accident_location	Indicates whether the accident happened in an urban or rural area	Used to compare urban and rural accident patterns
junction_control	Text	accident_conditions	Type of control at the junction, such as signals or give way signs	Used to study junction-related accident conditions
junction_detail	Text	accident_conditions	Details of the junction where the accident occurred	Used to analyze accident patterns by junction type
light_conditions	Text	accident_conditions	Lighting condition during the accident, such as daylight or darkness	Used to analyze accidents by visibility condition

Column Name	Suggested Data Type	Source Table / Category	Description	Use in Analysis
carriageway_hazards	Text	accident_conditions	Hazards present on the carriageway, if any	Used to understand road hazard impact
road_surface_conditions	Text	accident_conditions	Surface condition of the road, such as dry, wet, snow, or frost	Used to analyze road condition risk
road_type	Text	accident_conditions	Type of road, such as single carriageway, dual carriageway, or roundabout	Used to identify accident-prone road types
weather_conditions	Text	accident_conditions	Weather condition at the time of the accident	Used to analyze weather impact on accident frequency and severity
vehicle_type	Text	accident_conditions	Type of vehicle involved in the accident	Used to analyze casualty patterns by vehicle type

Derived Columns Explanation

Derived Column	Formula / Logic	Purpose
year	Extracted from accident_date	Used for yearly filters and comparison
month	Extracted from accident_date	Used for monthly trend analysis
month_number	Extracted from accident_date	Used to sort months in correct order
hour	Extracted from accident_time	Used for time-based analysis
severity_score	Slight = 1, Serious = 2, Fatal = 3	Converts accident severity into a numeric score
serious_fatal_flag	Serious or Fatal = 1, Slight = 0	Used to calculate Serious/Fatal Accident Rate

SQL Table Mapping

The cleaned dataset was divided into three final SQL tables.

1. accident_fact

This is the central fact table. It stores the main measurable accident details.

Columns included:

- accident_index
- accident_date
- accident_time
- day_of_week
- accident_severity
- number_of_casualties
- number_of_vehicles
- speed_limit
- year
- month
- month_number
- hour
- severity_score
- serious_fatal_flag

2. accident_location

This table stores accident location-related information.

Columns included:

- accident_index
- latitude
- longitude
- local_authority_district
- police_force
- urban_or_rural_area

3. accident_conditions

This table stores road, weather, light, vehicle, and condition-related information.

Columns included:

- accident_index
- junction_control
- junction_detail
- light_conditions
- carriageway_hazards
- road_surface_conditions
- road_type
- weather_conditions
- vehicle_type

Relationship Key

The common key used to connect all final SQL and Power BI tables is:

accident_index

Relationship Structure

accident_fact.accident_index = accident_location.accident_index

accident_fact.accident_index = accident_conditions.accident_index

Key Measures Created in Power BI

Measure Name	Description
Total Accidents	Counts the total number of unique accident records
Total Casualties	Sums the total number of casualties
Total Vehicles Involved	Sums the number of vehicles involved in accidents
Fatal Accidents	Counts accidents where severity is Fatal
SeriousFatalAccidents	Counts accidents marked as Serious or Fatal
SeriousFatalRate	Calculates the percentage of accidents that are Serious or Fatal
Avg Casualties per Accident	Calculates average casualties per accident

Business Use of the Dataset

This dataset supports analysis of:

- Accident severity patterns

- Casualty risk
- Monthly and yearly accident trends
- Road type risk
- Vehicle type involvement
- Weather and road surface impact
- Urban vs rural accident distribution
- Light condition and visibility-related accident patterns
- District-level accident concentration

Summary

The Road Accident dataset contains accident-level records that help identify road safety patterns and risk factors. The data dictionary defines each column used in the project and explains how the dataset supports Excel analysis, SQL querying, and Power BI dashboard creation.